

CASE 26

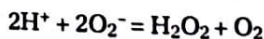
CHRONIC GRANULOMATOUS DISEASE

A specific failure of phagocytes to produce H_2O_2 and superoxide.

One of the most important mechanisms for destroying invading microbes in both the innate and adaptive immune responses is their phagocytosis by macrophages and neutrophils, after which they are killed rapidly within the phagocytic vacuoles, or phagosomes (Fig. 26.1). Uptake of microorganisms by phagocytes is enhanced by the opsonization of the particle—that is, coating it with complement, or, in the case of the adaptive immune response, with antibody and complement. The microbicidal actions of phagocytes occur through a variety of mechanisms. One of the most important of these involves the production of hydrogen peroxide (H_2O_2) and superoxide radicals (O_2^-), by a complex process that results in changes to the phagosomal pH and membrane potential, as well as the production of active bactericidal factors. These changes are critical in creating the bactericidal environment within the phagosome that facilitates the activation and function of the enzymes that are released into the phagosome. The enzyme NADPH oxidase catalyzes the initial reaction, which results in the generation of superoxide radicals:



The superoxide then undergoes dismutation to produce H_2O_2 :



NADPH oxidase is a large enzyme complex that is assembled in the phagosome membrane from component parts originally residing in the granule membranes

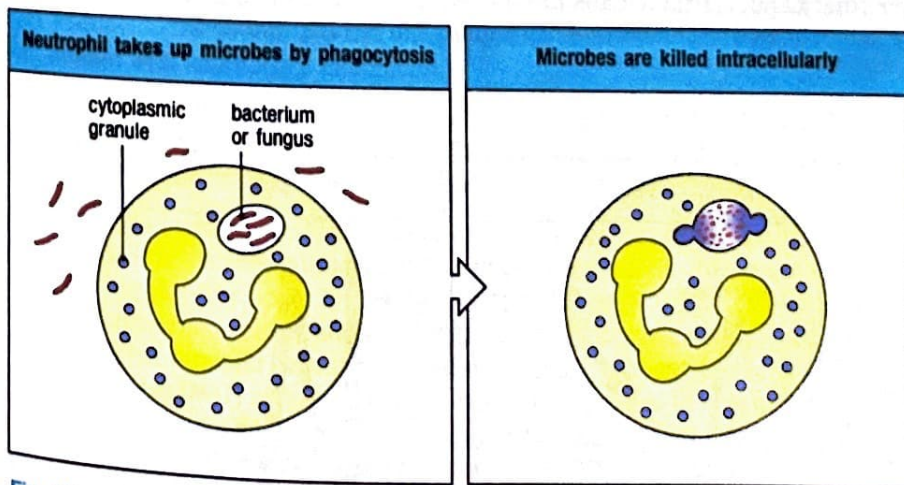


Fig. 26.1 Activation of a neutrophil. Microbes (red) are ingested by a phagocyte and enter the cytoplasm in a phagocytic vacuole, or phagosome. The phagosome fuses with a cytoplasmic granule containing microbicidal enzymes. The enzymes are released into the phagosome, where they kill and degrade the microbe.

TOPICS BEARING
ON THIS CASE:

- Microbicidal action of phagocytes
- Leukocyte migration
- Chronic inflammation

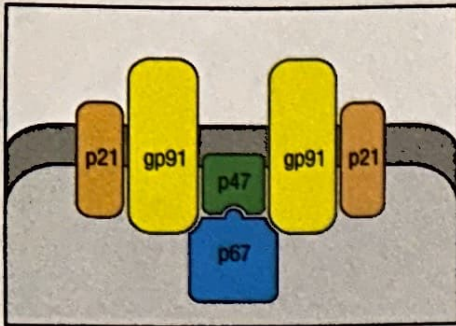


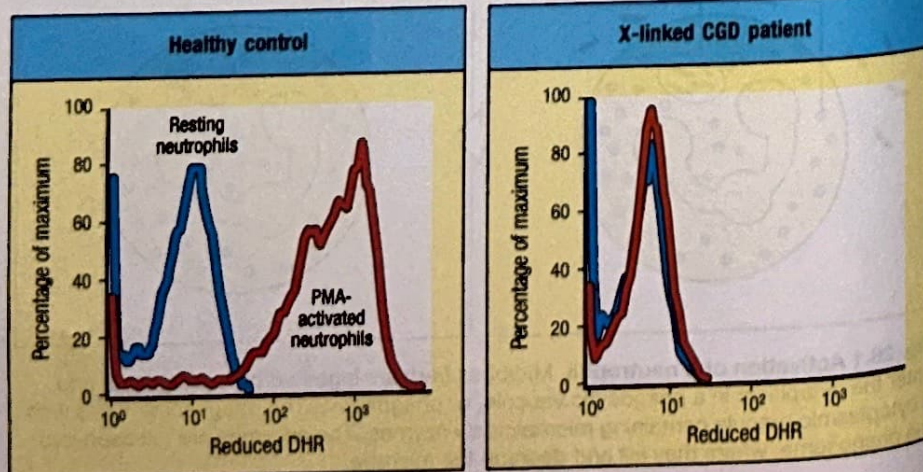
Fig. 26.2 The NADPH oxidase complex. NADPH is a large, multisubunit enzyme complex whose assembly in the phagosome membrane is triggered by a phagocytic stimulus. For simplicity, only four major components are shown here: gp91^{phox} and p22^{phox}, which together form the complex known as cytochrome b₅₅₈ in the granule membranes, and p67^{phox} and p47^{phox}, which normally reside in the cytoplasm. On delivery of a phagocytic stimulus to the cell, p47^{phox} becomes hyperphosphorylated and binds to p67^{phox} and other cytosolic components, and these components migrate to the phagosome membrane, where they assemble with cytochrome b₅₅₈ to form the complete NADPH oxidase.

and the cytosol. The membrane complex cytochrome b₅₅₈, which contains the catalytic site of NADPH oxidase, is made of a heavy chain (gp91^{phox}) and a light chain (p22^{phox}) (Fig. 26.2). Several other subunits, including p47^{phox}, p67^{phox}, p40^{phox}, and Rac2, reside in the cytoplasm in unstimulated phagocytes. When a phagocyte is stimulated, the cytosolic subunits translocate to the membrane where they interact with cytochrome b₅₅₈ to form active NADPH oxidase, which shuttles electrons from cytosolic NADPH to molecular oxygen (O₂). The genes encoding p47^{phox}, p67^{phox}, p21^{phox}, and p40^{phox} map to autosomal chromosomes, whereas gp91^{phox} is encoded on the short arm of the X chromosome. Defects in any of these components of the NADPH oxidase enzyme impede its function, and result in a disease with a common phenotype, called chronic granulomatous disease (CGD). The most common form of the disease is X-linked CGD, which is caused by mutations in the gene encoding gp91^{phox}.

A diagnosis of CGD can be ascertained using the dihydrorhodamine (DHR) flow cytometry assay to see if phagocytic cells have functional NADPH oxidase. In this method, whole blood is loaded with DHR, and the cells are then stimulated with phorbol 12-myristate 13-acetate (PMA). In normal subjects, PMA-activated neutrophils produce superoxide radicals that reduce DHR to the fluorescent molecule rhodamine. Flow cytometry is used to enumerate the proportion of polymorphonuclear cells that reduce DHR in response to activation with PMA (Fig. 26.3). This assay is able to identify patients with residual (but reduced) NADPH oxidase function due to hypomorphic mutations. The DHR assay can also identify carrier females of X-linked CGD. In these women, because of random X chromosome inactivation, two populations of neutrophils exist. A proportion of their neutrophils express the normal copy of the *CYBB* gene (encoding for gp91^{phox}), and are identified by a fluorescent peak in the DHR flow cytometry output. The other population of neutrophils express the X chromosome carrying the mutant form of *CYBB*, so these cells are unable to reduce DHR to rhodamine, and hence the cells do not fluoresce.

Until twenty years ago, reduction of nitro blue tetrazolium (NBT) was used as the diagnostic assay for CGD. To perform the NBT assay, a drop of blood suspended on a slide is treated with PMA and NBT. Neutrophils, which are the main phagocytes in blood, take up the NBT and the PMA. In normal neutrophils, the colorless NBT is reduced to a dark purple, insoluble formazan, which is easily seen in the phagocytic cells viewed under a microscope. In blood from patients with CGD, NBT is not reduced, so it remains colorless, and the cells do not turn purple (Fig. 26.4). Unlike the DHR flow cytometry assay, the NBT test is not quantitative, making it difficult to correctly identify patients with residual (but defective) NADPH oxidase activity. On the other hand, carrier females of X-linked CGD can be identified by the NBT test, because the two populations of neutrophils can be differentiated.

Fig. 26.3 DHR assay for analysis of NADPH oxidase activity in a healthy individual and a patient with X-linked CGD. Whole blood was stained with dihydrorhodamine and activated with phorbol 12-myristate 13-acetate (PMA). The samples were then analyzed by FACS, selecting neutrophils by their surface characteristics. Normal phagocytes produce superoxide radicals that reduce DHR to rhodamine, a highly fluorescent molecule. In normal individuals, almost 100% of the neutrophils become activated, causing reduction of the DHR, whereas in CGD patients the cells fail to reduce the DHR and hence remain non-fluorescent.



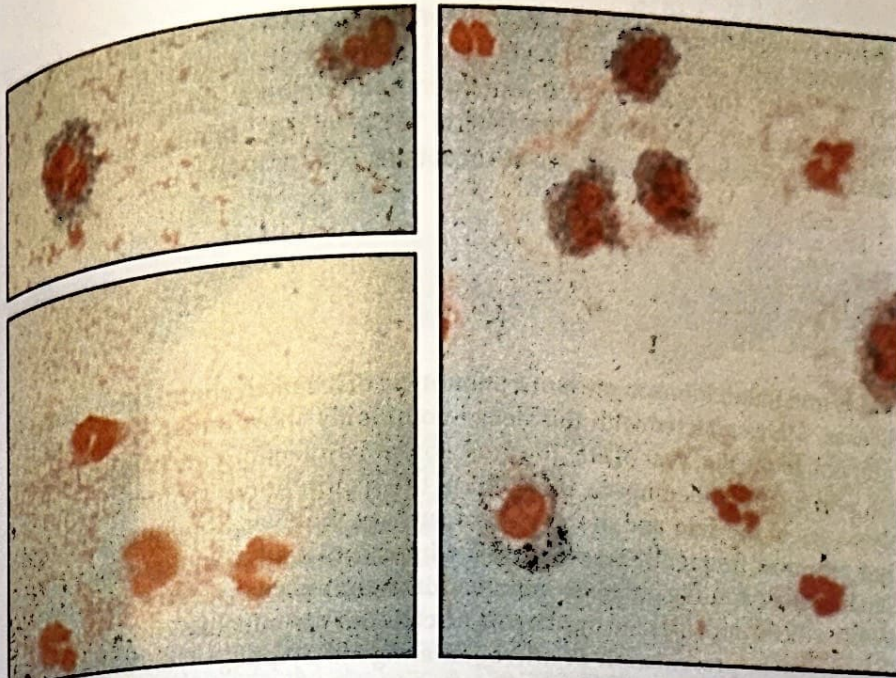


Fig. 26.4 Neutrophils stimulated with phorbol myristate acetate in the presence of nitro blue tetrazolium (NBT) dye. The upper left panel shows two normal neutrophils stained purple. The lower left panel shows neutrophils from a patient with CGD that failed to reduce NBT. The neutrophils in the right panel are from a heterozygous carrier of chronic granulomatous disease. Half the neutrophils have reduced NBT (purple), and half do not stain. Photographs courtesy of P. Newburger.

The case of Randy Johnson: a near death from exotic bacteria and fungi.

Randy Johnson, a 15-year-old high-school student, had a summer job working with a gardening crew. His job entailed spreading bark mulch in garden beds. At the end of August of that summer he rapidly developed severe shortness of breath, a persistent cough, and chest pain. Because of impending respiratory failure he was admitted to the hospital. A radiological examination of Randy's chest was performed, which revealed the presence of large 'cotton ball' densities in both lungs (Fig. 26.5). One of these lesions was aspirated with a fine needle; when the aspirate was stained, numerous fungal hyphae were seen. A culture of the aspirate grew *Aspergillus fumigatus*.

Randy was started on intravenous amphotericin B and assisted mechanical ventilation through a tracheotomy. He slowly improved over the course of two months. During this time in the hospital, he contracted two further respiratory infections, with *Pseudomonas aeruginosa* and *Streptococcus faecalis*. These infections were also treated with appropriate antibiotics.

At the time of his initial admission to the hospital, Randy's white blood cell count was $11,500 \mu\text{l}^{-1}$ (normal $5000\text{--}10,000 \mu\text{l}^{-1}$). He had 65% neutrophils, 30% lymphocytes, and 5% monocytes: these proportions are normal.

Because of his infections, which were unusual for a seemingly healthy 15-year-old adolescent, Randy's serum immunoglobulins were measured. The serum IgG level was 1650 mg dl^{-1} (the upper limit of normal is 1500 mg dl^{-1}). IgM and IgA were in the high normal range at 250 and 175 mg dl^{-1} , respectively. Because no defect could be found in his humoral immunity, his white cell function was tested with a nitro tetrazolium blue (NBT) slide test. His white blood cells failed to reduce NBT. Randy had four older sisters and two brothers. They all had an NBT test performed on their blood. One, his brother Ralph, aged 9 years, also failed to reduce NBT. His mother reported that Ralph had had a perirectal abscess in infancy but had otherwise been well. Randy's mother and one sister had a mixed population of neutrophils in the NBT test; about half of the phagocytes reduced NBT and the other half did not (see Fig. 26.4). His other three sisters and other brother exhibited normal NBT reduction.

Fifteen-year-old boy with impending respiratory failure.

Repeated respiratory infections.

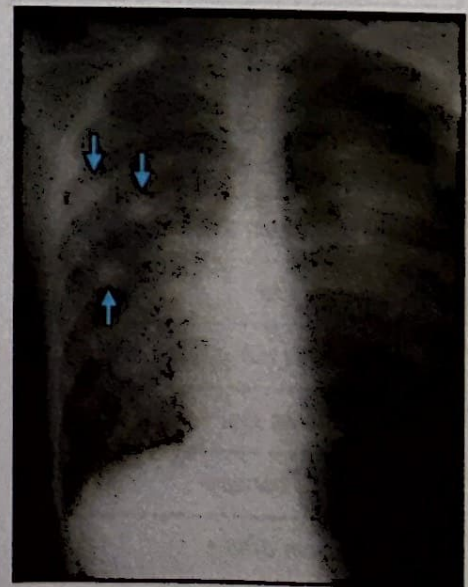


Fig. 26.5 A radiograph of *Aspergillus* pneumonia. Arrows point to 'cotton ball' densities characteristic of fungi.

Further studies of Randy's and Ralph's granulocytes revealed that the rate of H_2O_2 production was 0.16 and 0.028 nmol min^{-1} per 10^6 cells (normal 6.35 nmol min^{-1} per 10^6 cells). The cytochrome *b* content in both brothers' granulocytes was less than 1.0 pmol mg^{-1} protein (normal control 101 pmol mg^{-1} protein). Both Ralph and Randy were started on treatment with injections of interferon- γ .

Chronic granulomatous disease (CGD).

Randy and Ralph Johnson present a phenotype of X-linked chronic granulomatous disease. Males affected with this defect commonly show undue susceptibility to infection in the first year of life. The most common infections are bacterial and fungal, and include pneumonia, infection of the lymph nodes (lymphadenitis), and abscesses of the skin and of the viscera such as the liver. The fungi and bacteria that are most frequently responsible for infections and granuloma formation in CGD are listed in Fig. 26.6. Granuloma formation occurs when microorganisms are opsonized and ingested by phagocytic cells but the infection cannot be cleared. The persistent presentation of microbial antigens by phagocytes induces a sustained cell-mediated response by $CD4^+$ T cells, which recruit other inflammatory cells and set up a chronic local inflammation called a granuloma. This can occur in normal individuals in response to intracellular bacteria that colonize macrophages (Fig. 26.7); but in CGD the intracellular microbicidal mechanisms are defective, so many infections persist, leading to chronic inflammatory reactions resulting in granulomas. Furthermore, granulomatous lesions may occur independent of an infection. They preferentially affect hollow organs (e.g., gastrointestinal tract, urinary tract), causing strictures and other complications. Inflammatory bowel disease has also been frequently associated with CGD.

A correlation has been reported in patients with CGD between NADPH oxidase activity and the risk of and survival from serious infectious and inflammatory complications. In particular, patients with no NADPH oxidase activity have a more severe prognosis. Continuous antimicrobial prophylaxis is the primary treatment for CGD. Trimethoprim-sulfamethoxazole is used commonly to prevent bacterial infections. Itraconazole is used for anti-fungal prophylaxis. Patients with CGD should avoid exposure to significant sources of *Aspergillus* (e.g., hay, mulch, rotting wood, compost piles). Patients should also be actively immunized to reduce the risk of infections.

Acute infections require a thorough work-up, with the goal of identifying the pathogen and its sensitivity to drugs. Anti-inflammatory medications are often required to treat inflammatory disease. Steroids may be beneficial in these cases; furthermore, a low dose of steroids is often used during acute infections, especially if severe inflammatory responses (thick granulomatous lesions) may impede penetration of antibiotics into the lesion. Regular administration of Interferon- γ improves the resistance to infection of males with X-linked CGD. The basis for this effect is not fully understood.

Hematopoietic stem cell transplantation (HSCT) is currently considered the only curative treatment for CGD. HSCT replaces the defective blood-cell progenitors of the patient with normal progenitors from a donor. Significant improvement in outcomes has been reported recently with the use of a submyeloablative conditioning regimen during HSCT for CGD. This approach permits robust and sustained engraftment of donor cells in transplanted patients, thereby curing the disease.

Gene therapy has been attempted in patients with X-linked CGD. Hematopoietic stem cells from the patients have been transduced with a retroviral vector encoding for $gp91^{\text{phox}}$, and then infused back into the patient. Use of chemotherapy has facilitated engraftment of donor derived cells; however, clonal expansion of myeloid cells in which the vector had inserted near proto-oncogenes, has been observed,

<i>Staphylococcus aureus</i>
<i>Aspergillus</i> spp.
<i>Chromobacterium violaceum</i>
<i>Pseudomonas cepacia</i>
<i>Nocardia</i> spp.
<i>Salmonella typhimurium</i>
<i>Serratia marcescens</i>
<i>Mycobacterium fortuitum</i>
<i>Klebsiella</i> spp.
<i>Escherichia coli</i>
<i>Actinomyces</i> spp.
<i>Legionella bosmanii</i>
<i>Clostridium difficile</i>

Fig. 26.6 Table of fungal and bacterial infectious agents most commonly responsible for infections in CGD.

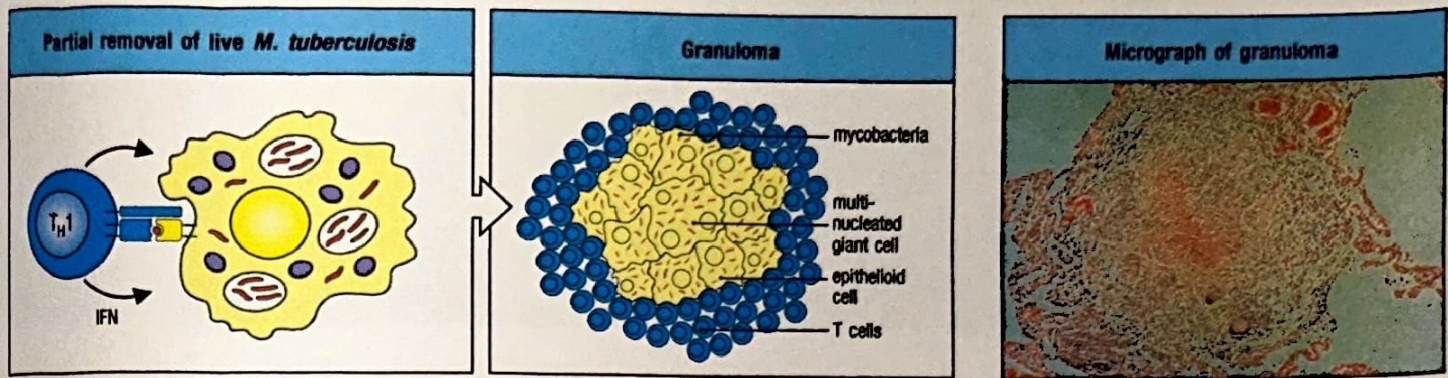


Fig. 26.7 Granulomas form when an intracellular pathogen or its constituents cannot be eliminated completely. The pathogen illustrated here is the intracellular bacterium *Mycobacterium tuberculosis*. In the left panel, mycobacteria (red) are shown resisting the effects of macrophage activation by a T_H1 $CD4^+$ T cell. The middle panel is a schematic diagram of a granuloma, which forms as the result of a localized inflammatory response that develops because of the sustained activation of chronically infected phagocytes. The granuloma

consists of a central core of infected macrophages, and may include multinucleate giant cells, which are fused macrophages, surrounded by large macrophages often called epithelioid cells. These may, in turn, be surrounded by T cells, many of which are $CD4^+$. Granulomas frequently form in CGD because macrophages are unable to destroy the bacteria they ingest, and infected macrophages accumulate. The right panel shows a micrograph of a granuloma. Photograph courtesy of J. Orrell.

raising serious safety concerns. Furthermore, silencing of the $gp91^{phox}$ transgene after few months has been associated with disappearance of clinical benefit. More recently, modified self-inactivating lentiviral vectors have been generated, and their safety and efficacy are currently being tested in clinical trials.

Questions.

- 1 In Fig. 26.4, 50% of Randy's mother's neutrophils have reduced the NBT and 50% have not. How do you explain this?
- 2 Children with CGD do not have problems with pneumococcal infections. Could you hypothesize why this is so?
- 3 The immunoglobulin levels in Randy's blood were somewhat elevated. How do you explain this?
- 4 A child presents with a clinical picture of CGD. In addition to defiant neutrophil superoxide production, her neutrophils exhibit defective chemotaxis. What gene in the membrane cytochrome b_{558} complex would you suspect may be deficient?